Dataset Distillation Using Two Techniques

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Introduction

The scale of data in Machine learning continues to increase these days, along with the complexity of deep learning models. A new technique is gaining importance as the demand for reducing energy costs and shortening training times continues to grow.

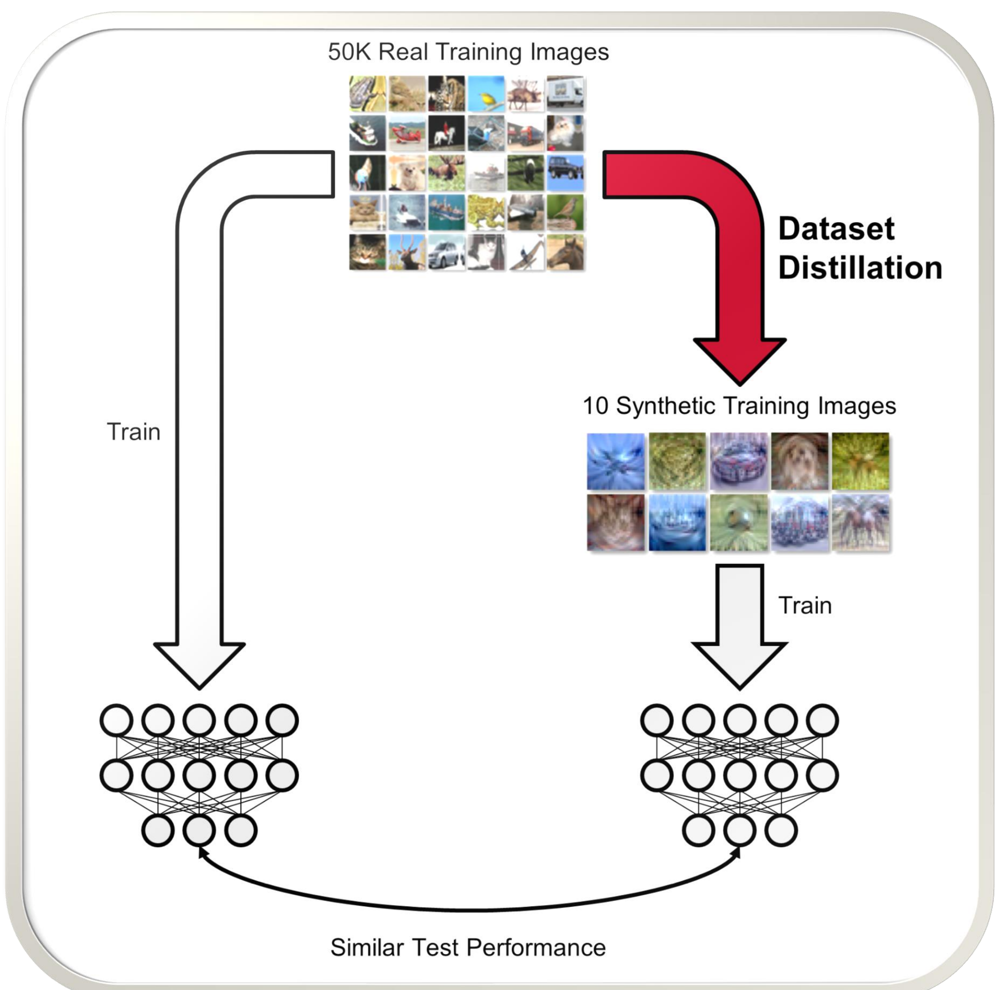


Fig. 1 Dataset Distillation Technique [1]

Dataset distillation is a machine learning approach aimed at identifying a representative subset of samples from a larger dataset, ensuring that this subset preserves the core information and statistical characteristics of the original dataset [2]. This technology holds significant promise for improving the scalability of machine learning systems in real-world settings.

In this poster, **Two** dataset distillation approaches are introduced.

- Dataset Distillation with Attention Matching (**DataDAM**): This methodology introduces an **attention-matching** algorithm, which aims to improve the model by training on a distilled dataset [3].
- Prioritized Alignment in Dataset Distillation (**PAD**): This methodology introduces the **extraction and embedding** techniques in the distillation process to refine data selection and preserve essential information [4].

Two pre-labeled databases are trained in this project, **MNIST and MHIST** [5,6].

Tab. 1 Properties of Databases

Dataset	Size	Shape	Class	Complexity	Domain
MNIST	70,000	(28,28,1)	10	Low	Handwritten
MHIST	3,152	(224,224,3)	Binary	High	Histopathology

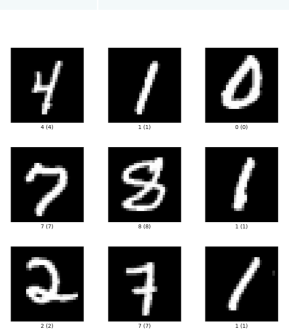


Fig. 2(a) MNIST [5]

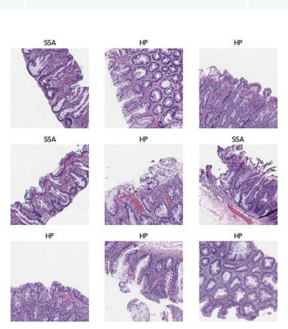


Fig. 2(b) MHIST [6]

Methods

General Steps

This project aim to test both approaches accuracy in the same parameter. It compares accuracies of original datasets and synthetic datasets. It also compares accuracies of two synthetic datasets using different methods and models.

Method
- DataDAM
- PAD

Database
- MNIST
- MHIST (DataDAM)

Model
- ConvNet
- LeNet

Start Set
- Random Select
- Gaussian Noise

Tab. 2 Important Hyper-parameter Values

Method	DataDAM	PAD	DataDAM
Dataset	MNIST	MNIST	MHIST
Optimizer	SGD	SGD	SGD
Batch Size	256	256	128
IPC	100	100	50
Random Weight	100	100	200

Synthetic Set Generation

By implementing the codebase of both paper, the synthetic sets are generated, and part of the sets are shown below. We mainly focus on the sets generated from gaussian noise using ConvNet.

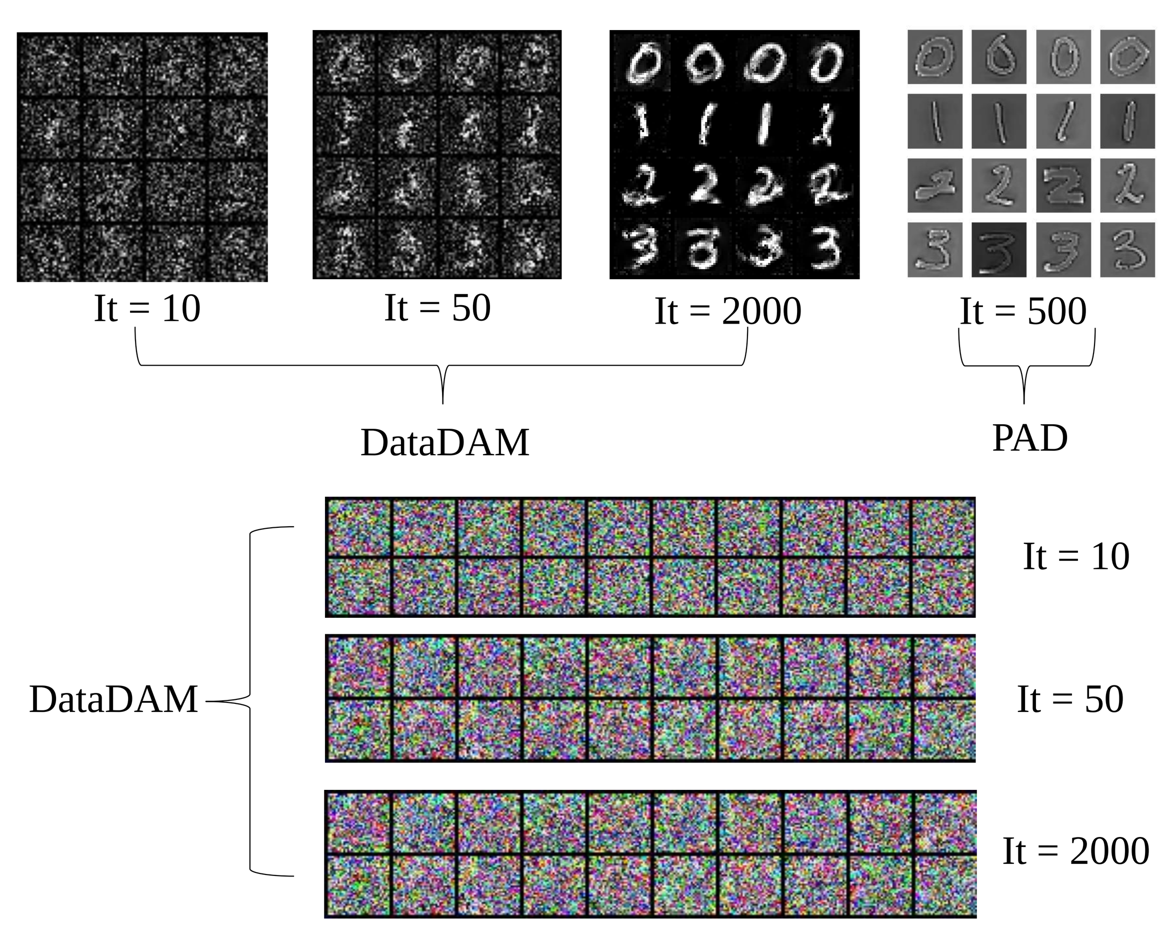


Fig. 2 Synthetic Datasets Evolution Visualization

In DataDAM, The MNIST image initialized by gaussian noise is hardly recognizable at low iterations. As iterations go larger, the images set starts to become recognizable. In PAD, the embedding phase successfully eliminated low information area.

The MHIST dataset initialized with noise using DataDAM method is hardly recognizable even at a large iteration.

Results

Original & Synthetic

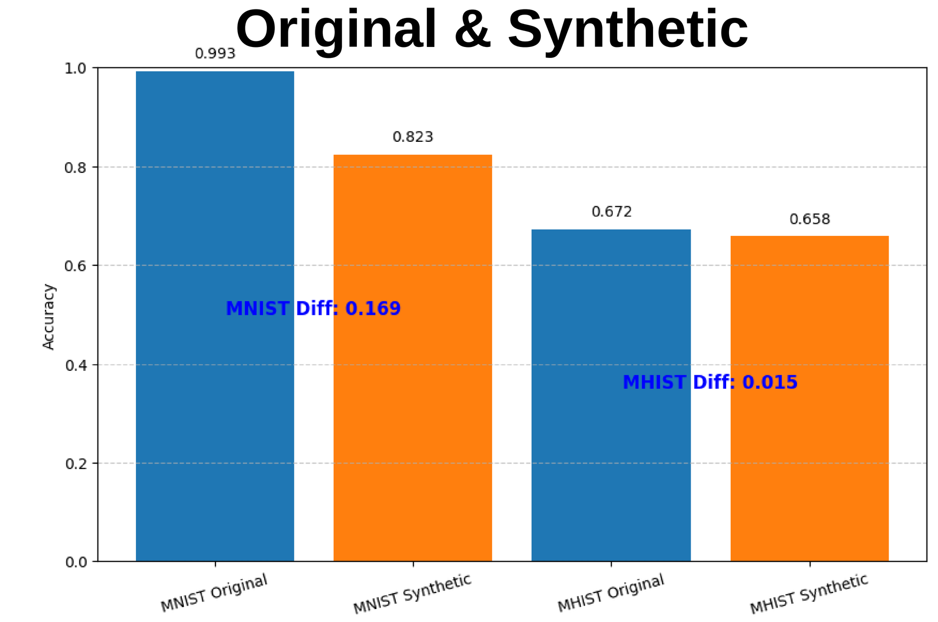
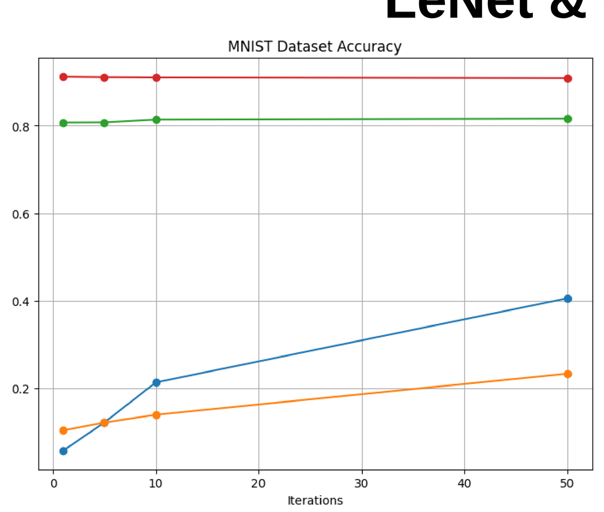


Fig. 3 Original and Synthetic Dataset Accuracy Result

For DataDAM, the comparison between original and synthetic set using two databases in iteration **10** are illustrated on the right. MNIST has a higher accuracy than MHIST. MNIST original and synthetic dataset has an accuracy difference of **0.169**. The MHIST original and synthetic dataset has an accuracy difference of **0.015**.

LeNet & ConvNet



The number of iterations has no or minimal impact on the accuracy of synthetic datasets initialized from random images. In contrast, for synthetic datasets initialized from Gaussian noise, accuracy improves progressively with an increase in the number of iterations.

- For MNIST, LeNet outperforms ConvNet in real data, ConvNet has better accuracy in noisy data testing than LeNet in larger iterations.
- For MHIST, accuracies of real data are similar for both models, but LeNet outperforms ConvNet in noisy data.

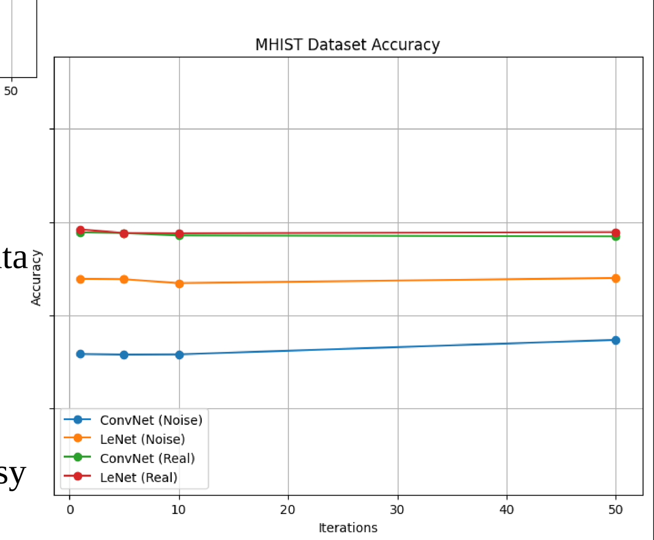


Fig. 4 Model Accuracies Comparison

DataDAM & PAD

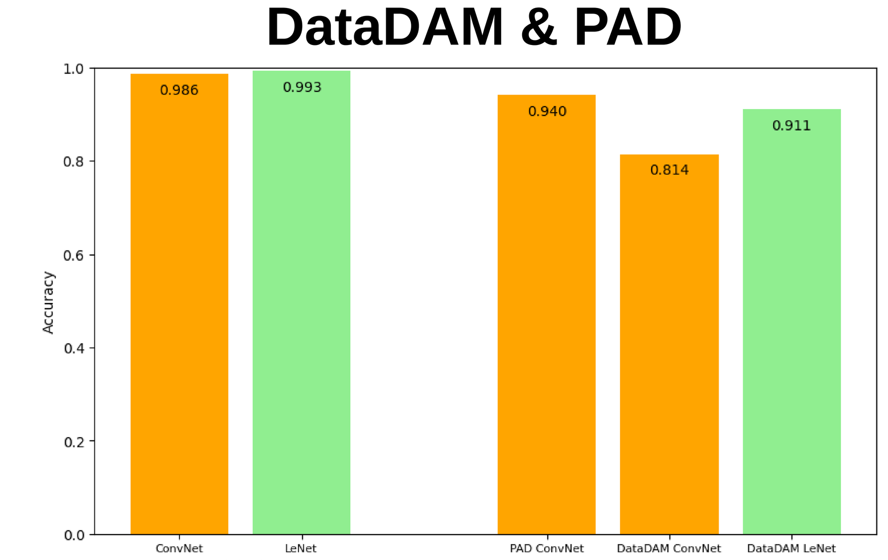


Fig. 5 MNIST Dataset Method Comparison

Both models perform well on original MNIST dataset. Comparing the DataDAM and PAD methods in same network (ConvNet), PAD is **12.6%** more accurate than DataDAM.

Conclusions

Synthetic Datasets:
- Synthetic datasets initialized from **random samples** maintain **stable accuracy** across iterations with minimal changes.
- Synthetic datasets initialized from **gaussian noise** exhibit progressive **accuracy improvements** as iterations increase, showing that iterative refinement is effective.
- Iterative processes are effective and are necessary for improving gaussian noise synthetic datasets.

Models:
- ConvNet is often more **complex** compared to LeNet. ConvNet model is harder to configure, and it demonstrates a **stronger learning capability** compared to LeNet.

Datasets:
- MNIST is a simpler dataset, requiring less feature extraction and achieving higher accuracy compared to MHIST. And MHIST is more complex and needs richer feature extraction capabilities.

Methods:
- Both methods contribute valuable strategies for dataset distillation.
- PAD demonstrates a strong performance in high IPC level than DataDAM.

Bibliography

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